Building a Beam-Class LLM

Reflection AI Beam 1.0: full technical analysis, competitive benchmarking against Anthropic, OpenAI, Google, xAI, Meta and open-weight leaders, and a business case for developing a competing model

Research baseline · Version 2.0

Item	Detail
Prepared for	Yossi Avni
Version / date	2.0, 19 October 2026 (supersedes v1.0 of 14 October 2026)
Information cut-off	Public sources available to 18–19 October 2026
Companion deliverable	Beam_Executive_Summary.pptx (executive deck)
What changed in v2.0	Beam weights released 16 Oct (Apache 2.0); model-card architecture confirmed (256 experts, top-8, 152K vocab, SWA-512); first independent score (Artificial Analysis Index = 30); xAI Grok, Meta Muse, Claude Opus/Sonnet 5.5, GPT-6, Gemini 4 added; market, pricing, business case, cost model, governance and risk sections added
Confidence convention	[V] vendor-reported · [I] independently measured · [E] author estimate · [S] single secondary source

All benchmark figures are as published by the cited sources and were not re-run by the author. Several frontier vendors report on different benchmark versions and harnesses; cross-vendor differences under ~2 points should be treated as noise.

Contents

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1. Executive summary

1.1 The bottom line

Reflection AI's Beam 1.0 is a 501B-parameter sparse Mixture-of-Experts (MoE) model with 23.4B active parameters per token, released with open weights under Apache 2.0 on 16 October 2026. It is the most capable American open-weight model released from a start-up to date, built for coding and agentic work, and it is unusually efficient: it ships with a 512-token sliding-window attention design that keeps a full-length KV cache in only 9 of its 52 layers, and it uses ~40% fewer output tokens than comparable models on the Artificial Analysis test suite.

However, Beam is not a frontier model. On the first independent evaluation (Artificial Analysis Intelligence Index v4.3.x) it scores 30, versus 44–46 for the best Chinese open-weight models (Kimi K3, GLM-5.3, DeepSeek V4.1, MiMo-V2.6), 46 for xAI's Grok 4.7, 52.6–53 for Google's Gemini 4 Argon and OpenAI's GPT-6 Astra, and 56–58 for Anthropic's Claude Sonnet 5.5 and Opus 5.5. Reflection's own launch table (parity with GLM-5.2 on reasoning) is therefore optimistic relative to independent measurement.

1.2 What this means for building a competing model

Ambition level	Independent target	What it takes [E]	Indicative 2-yr cost [E]
1. Beam parity	AA Index ≈ 30; Beam KPIs in Section 11	Adapt Beam (Apache 2.0) or train a ~500B-A23B MoE on ~24T tokens + large RL	≈ US$10M (adapt) to ≈ US$175M (from scratch)
2. Open-frontier parity	AA ≈ 45 (Kimi/GLM/DeepSeek class)	~1–2T total, 40–60B active, 30T+ tokens, RL compute ≥ pretraining	≈ US$300–500M
3. Closed-frontier challenger	AA ≥ 53 (GPT-6 / Gemini 4 / Claude class)	Multi-trillion-parameter models, 10^25–10^26 FLOPs, large proprietary data and RL environment fleets	US$1B+ per generation; only realistic for hyperscaler-backed labs

1.3 Recommendation

1. **Do not attempt to out-compete Anthropic, OpenAI, Google or xAI head-on with a Beam-class model.** The independent gap is ~23–28 index points, and closed vendors hold ~89% of enterprise API spend.
2. **Exploit the open-weight opportunity Beam has created.** Beam's Apache 2.0 weights, reference implementation, Mirror CLI training harness and day-0 vLLM/SGLang support lower the entry cost for a specialised or sovereign model by one to two orders of magnitude.
3. **Pursue a staged plan:** Stage 1 (0–6 months, ≈US$10M): adapt Beam for a chosen vertical/sovereign niche and prove the data, RL and evaluation stack. Stage 2 (6–18 months, gated): continued pre-training and full post-training (≈US$55M). Stage 3 (gated on traction and capital): a from-scratch next-generation model.
4. **Compete on what closed vendors cannot offer:** on-premises/sovereign deployment, data residency, full customisation, predictable cost per token, and domain specialisation — the segments where open weights win despite a lower aggregate index.

1.4 Key numbers

Metric	Value	Basis
Beam total / active parameters	501.3B / 23.4B	[V] model card
Experts / active per token	256 / 8 (+3 dense layers)	[V] model card, vLLM recipe
Pretraining tokens / GPUs	23.8T tokens / 6,144 GB300, < 4 weeks	[V]
RL compute	10,500 GB300 × ~4 weeks; >100M rollouts	[V]
SWE-bench Verified (BF16)	81.6%	[V] model card
Artificial Analysis Intelligence Index	30 (19th of 120)	[I]
Leader on same index	Claude Opus 5.5 (Max) = 58	[I]
Open-weight share of enterprise API spend	11% (down from 19%)	[S] Menlo Ventures, late 2025
Estimated Beam training compute	≈ 3.3 × 10^24 FLOPs pretraining; ≈ 11–14M GPU-hours in total	[E]

2. Purpose, scope and methodology

This document is the technical and commercial baseline for a programme to research, design and build a new large language model with capabilities equivalent to — and ideally exceeding — Reflection AI's Beam. It is written for the founding technical team, investors and partners.

2.1 Questions answered

- What exactly is Beam (architecture, data, training, release package) and what remains undisclosed?
- How does Beam perform against Anthropic, OpenAI, Google, xAI, Meta and the leading open-weight models — on vendor-reported and on independent benchmarks?
- What KPIs define "Beam-equivalent", and what would it take to reach higher tiers?
- What are the strategic options, costs, team, infrastructure, risks and go-to-market paths for a new model?
- Which research directions should shape the design?

2.2 Sources and confidence

Primary sources: Reflection's launch and availability posts, the Hugging Face model cards (BF16, FP8, NVFP4), the Hugging Face, AWS, Google Cloud, NVIDIA and vLLM integration pages, and vendor system cards. Independent sources: Artificial Analysis, Vals AI and public leaderboards. Secondary sources: technical press and analyst blogs. Direct page retrieval was restricted in the research environment, so some figures were taken from search-indexed extracts of these sources; each is tagged with a confidence label (see document control).

2.3 Known limitations

- Beam's full technical report is still marked "coming"; hidden size, optimiser and data mixture are not yet public.
- Closed vendors publish different benchmark versions (e.g., Terminal-Bench 4.0 vs 2.1; SWE-bench Pro vs Verified). Only the Artificial Analysis index puts all models on one scale.
- Market-size estimates vary widely by definition; they are used directionally only.

3. Reflection AI: company profile

Item	Detail
Founded	2024, by former Google DeepMind researchers
Leadership	Misha Laskin (CEO); Ioannis Antonoglou (CTO, co-creator of AlphaGo)
Mission	An American open frontier lab producing open-weight models competitive with leading Chinese open models
Funding	US$2B (Oct 2025) led by NVIDIA (~US$800M) at US$8B valuation; reported talks in Mar 2026 for US$2.5B at ~US$25B pre-money; one aggregator lists US$27.5B (May 2026) [S]
Products	Asimov (code-research agent); Beam 1.0 (open weights); Mirror CLI (open-source agent harness used to train Beam)
Ecosystem	40+ launch partners incl. AWS, Google Cloud, Microsoft Azure, NVIDIA, Hugging Face, Baseten, Ollama, LM Studio
Strategic news	Reports (Oct 2026) of early NVIDIA talks to acquire, invest further in, or license technology from Reflection [S]

3.1 Timeline

Date	Event
2024	Company founded
Oct 2025	US$2B round led by NVIDIA
Mar–May 2026	Reported raise at US$25–27.5B valuation [S]
5 Oct 2026	Beam announced: specs, launch benchmark table, waitlisted beta API (262K context)
16 Oct 2026	Weights released on Hugging Face (BF16, FP8, NVFP4) with tokenizer, reference implementation, DFlash2 drafter and Mirror CLI; day-0 support in Transformers, vLLM and SGLang; Beam Safety Technical Report published
Oct 2026	Artificial Analysis scores Beam 30 on Intelligence Index v4.3.x
Pending	Full Beam technical report

Not to be confused with "Reflection Llama-3.1 70B" (2024), an unrelated fine-tune by a different team.

4. Beam 1.0: complete specification

Attribute	Value	Source
Official name	Beam-1.0-501B-A23B	[V] Hugging Face
Type	Decoder-only sparse MoE transformer, text-only	[V]
Total / active parameters	501.3B / 23.4B (≈4.7% active)	[V] model card
Layers	52 = 3 dense-MLP layers + 49 MoE layers	[V] model card
Experts	256 routed experts per MoE layer; top-8 routing	[V] model card, vLLM recipe
Attention	Gated GQA; 512-token sliding-window attention on most layers; full-attention layers use NoPE (no positional encoding)	[V] card; [S] vLLM recipe
Full-attention layers	9 of 52 keep a full-length KV cache	[S] GGUF quantiser notes
Normalisation / residual	SandwichNorm; depth-based residual scaling; FP32 residual accumulation	[V] launch post
Load balancing	Auxiliary-loss-free expert-bias balancing with cosine-decayed bias updates; busiest expert ≈1.04× mean	[V] launch post
Vocabulary	152K tokens	[V] model card
Context length	1M tokens (262K on the beta API)	[V]
Pretraining	23.8T tokens; 6,144 GB300 GPUs; < 4 weeks	[V]
Data curation	~95% of raw web tokens discarded	[S] MarkTechPost
RL post-training	10,500 GB300 GPUs; ~4 weeks; >100M rollouts; ~1M environments; ~1.3B sandboxes; 256K max rollout context; Mirror CLI harness	[V]
Release formats	BF16, FP8, NVFP4 checkpoints; DFlash2 speculative-decoding drafter (~1.93B params)	[V]; drafter size [S]
License	Apache 2.0 (weights, tokenizer, reference implementation)	[V]
Hidden size, heads, optimiser, data mixture	Not yet disclosed	—

5. Architecture deep-dive

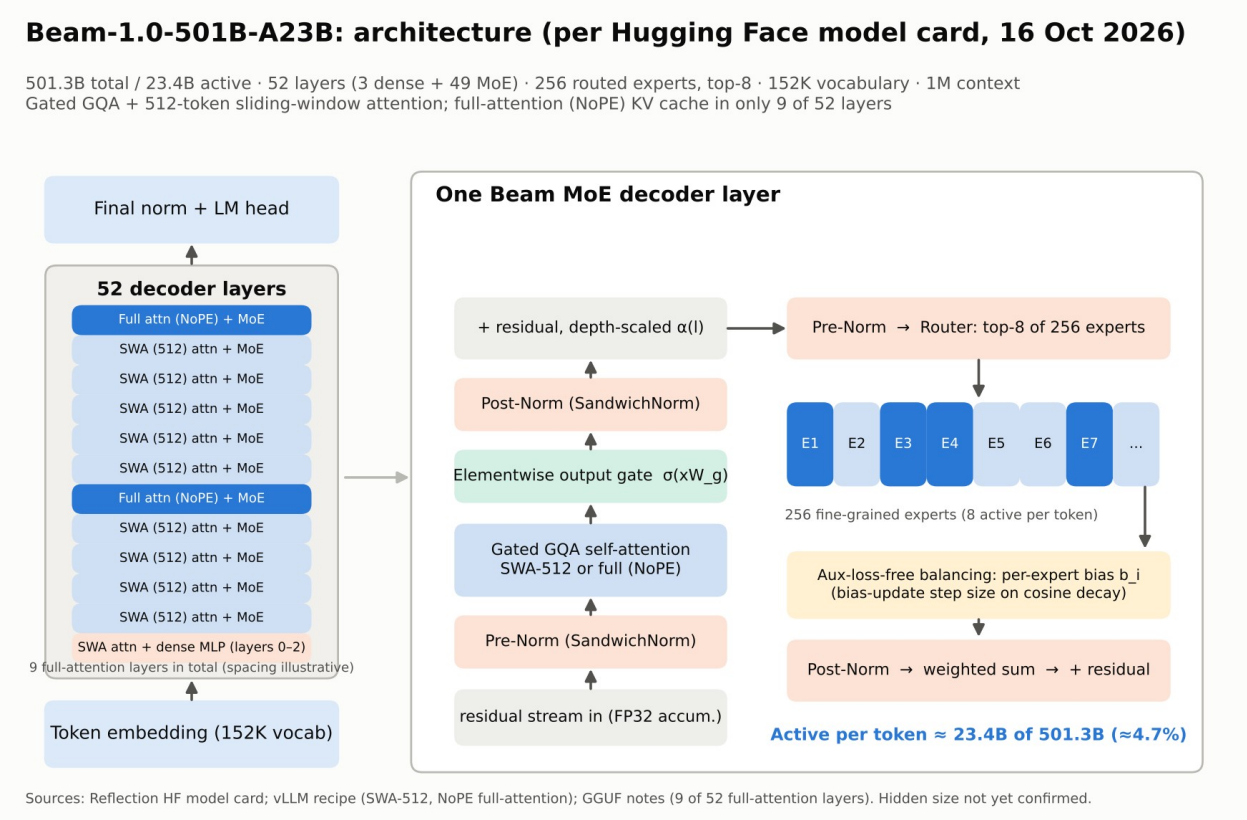


Figure 1 — Beam 1.0 architecture as confirmed by the model card and serving recipes.

5.1 Sparse MoE backbone: 256 experts, top-8

The first three layers use dense MLPs (a common choice for stabilising early representations); the remaining 49 layers replace the MLP with 256 fine-grained routed experts, of which a learned router selects 8 per token. With 8/256 experts active, each token touches ~3% of expert parameters; together with attention and embeddings this yields 23.4B active parameters. This configuration closely follows the DeepSeek-V3 lineage (256 routed experts, top-8), now the de-facto standard for open MoE models.

5.2 Auxiliary-loss-free balancing with cosine-decayed bias updates

Each expert has a bias added to its routing score for top-k selection only. Under-loaded experts have their bias raised after each step, over-loaded experts lowered. Reflection decays the bias-update step on a cosine schedule so routing settles late in training. Reported result: the busiest expert averages 1.04× the mean load — near-perfect utilisation, which directly improves expert-parallel throughput and avoids token dropping.

5.3 Attention: 512-token sliding window + sparse NoPE global layers

Most layers attend only to the previous 512 tokens, so their KV cache is constant regardless of context length. Only 9 of the 52 layers attend over the full sequence, and these use no positional encoding (NoPE), relying on the local layers to carry order information — a design that has been shown to extrapolate well to long contexts. Grouped-query attention (GQA) further shrinks KV size, and an elementwise output gate suppresses attention sinks.

KV-cache illustration at 1M tokens [E]	All 52 layers global	Beam (9 global + 43 SWA-512)
Relative KV memory	100%	≈ 17% (9/52; SWA layers negligible)
Relative attention FLOPs at long context	100%	≈ 17–20%
Practical effect	Long contexts need many GPUs for cache	1M context feasible on a single 8-GPU node in FP8/NVFP4

5.4 Stability components

- **SandwichNorm:** normalisation before and after each sub-layer to bound activation growth.
- **Gated attention output:** a learned sigmoid gate on attention outputs; reduces loss spikes and attention sinks.
- **Depth-based residual scaling:** scales residual-branch contributions by depth for stable signal propagation.
- **FP32 residual accumulation:** keeps the residual stream in FP32 while matrix multiplies run in low precision.

5.5 Inference accelerators

Beam ships with DFlash2, a small (~1.93B) speculative-decoding drafter that proposes several tokens for the main model to verify in parallel, targeted at the latency of agentic loops. Reflection's recipes recommend it for both vLLM and SGLang. Google Cloud reports day-0 serving with up to 5× performance improvement on its stack [V].

6. Training recipe

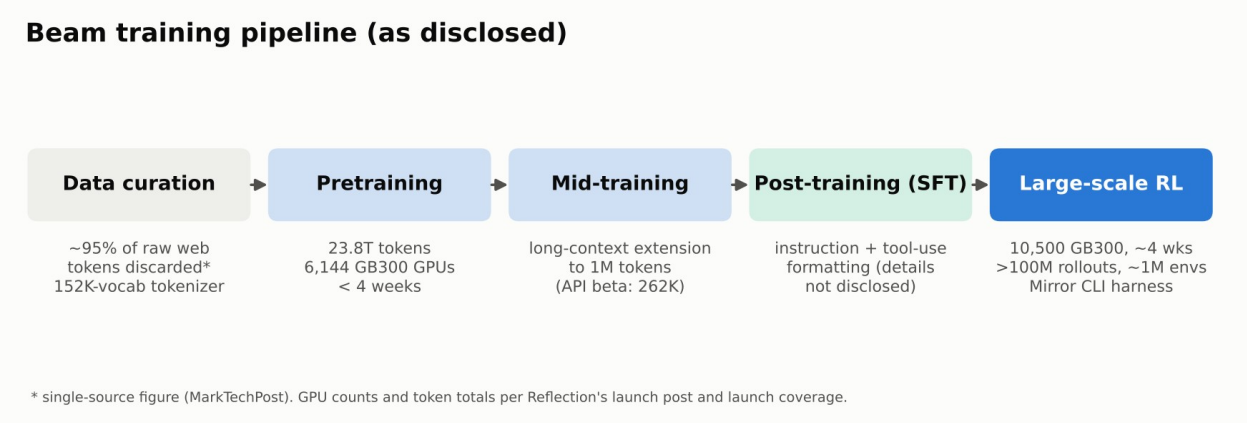


Figure 2 — Beam's disclosed training pipeline.

Phase	Disclosed facts	Implication for a replication
Data curation	~95% of raw web discarded; 152K-vocab tokenizer	Build model-based quality filters, deduplication, decontamination and synthetic code/maths before pre-training
Pre-training	23.8T tokens; 6,144 GB300; < 4 weeks	≈ 3.3 × 10^24 FLOPs (6 × 23.4B × 23.8T) [E]; requires ~20–35% MoE MFU
Mid-training	Context extended to 1M tokens	Long-document and repository-level data; staged sequence-length increase
SFT	Not detailed	Seed tool-use formats, multi-turn agent traces and reasoning style
RL	10,500 GB300 × ~4 weeks; >100M rollouts; ~1M environments; ~1.3B sandboxes; Mirror CLI	Asynchronous RL, sandbox orchestration and verifiable rewards are core engineering, not afterthoughts

6.1 Mirror CLI: the harness Beam was trained against

Reflection open-sourced Mirror CLI, the terminal-agent harness used during RL (reported as a Python front-end for prompts, skills and tooling with a Rust agent core). Training against a production harness means the model's agentic behaviour matches how it will be deployed. For a new model, adopting Mirror CLI (or an equivalent) as both the RL environment interface and the evaluation harness removes a large source of train/serve mismatch.

6.2 Compute split

In GPU-hours Beam's RL phase (≈7.1M GB300-hours) was roughly 2–3× its pre-training phase (≈2.5–3M) [E]. This inverts the historical ratio and reflects the 2026 consensus that agentic capability is mostly bought in post-training.

7. Release package, deployment and hardware

7.1 Checkpoints and quality by precision (model card)

Benchmark	BF16	FP8	NVFP4
SWE-bench Verified	81.6	80.2	80.7
SWE-bench Pro v1	66.1	65.9	65.6
Terminal-Bench 2.1	80.1	78.9	78.4
τ^3-bench [S]	36.1	36.8	35.4

Quantisation to FP8/NVFP4 costs ≤ 1.7 points on these agentic benchmarks — a strong argument for NVFP4-first serving.

7.2 Memory footprint and minimum hardware [E]

Format	Weights size (approx.)	Minimum practical deployment
BF16	≈ 1.0 TB (≈1,093 GB incl. overhead per one sizing tool)	16× H200/B200 or 8× GB300/B300 (288 GB each)
FP8	≈ 0.5 TB	8× H200 (141 GB) or 8× B200/B300
NVFP4	≈ 0.28 TB	4–8× Blackwell-class GPUs
Community GGUF (Q4)	≈ 0.28–0.3 TB	Large-memory workstations / CPU+GPU offload (unsloth, AtomicChat quantisations)

7.3 Where Beam runs today

- Frameworks: Hugging Face Transformers, vLLM and SGLang (day-0), NVIDIA NeMo AutoModel and Dynamo recipes.
- Clouds and platforms: AWS, Google Cloud, Microsoft Azure, NVIDIA, Baseten, and 40+ partners; local via Ollama and LM Studio.
- Pricing: no Reflection rate card; partner pricing varies (Baseten listed at launch). Artificial Analysis had not yet measured speed or price.

8. Performance analysis

8.1 Model scale in context

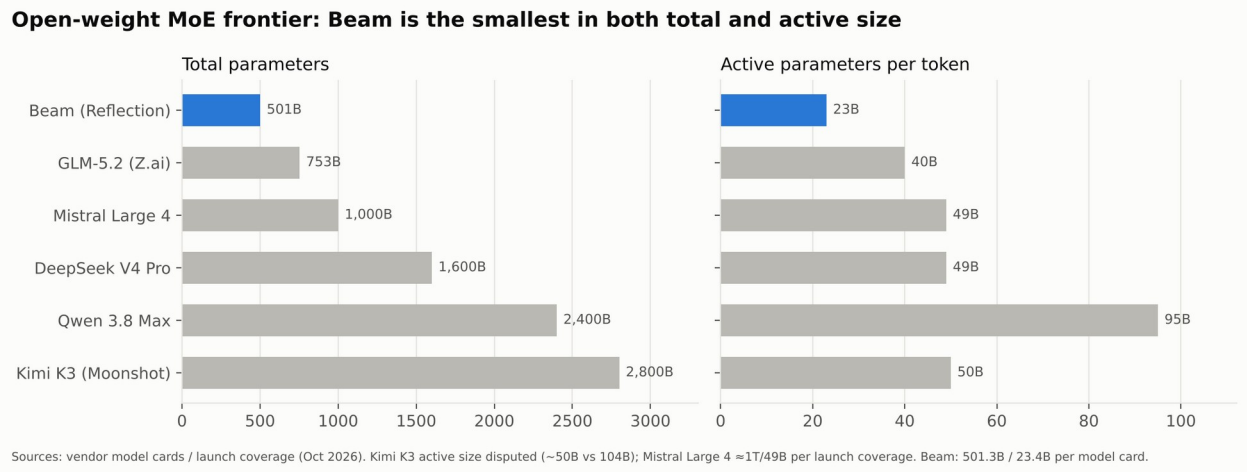


Figure 3 — Total and active parameters of Beam and open-weight competitors.

8.2 Vendor-reported: Beam vs. open-weight peers

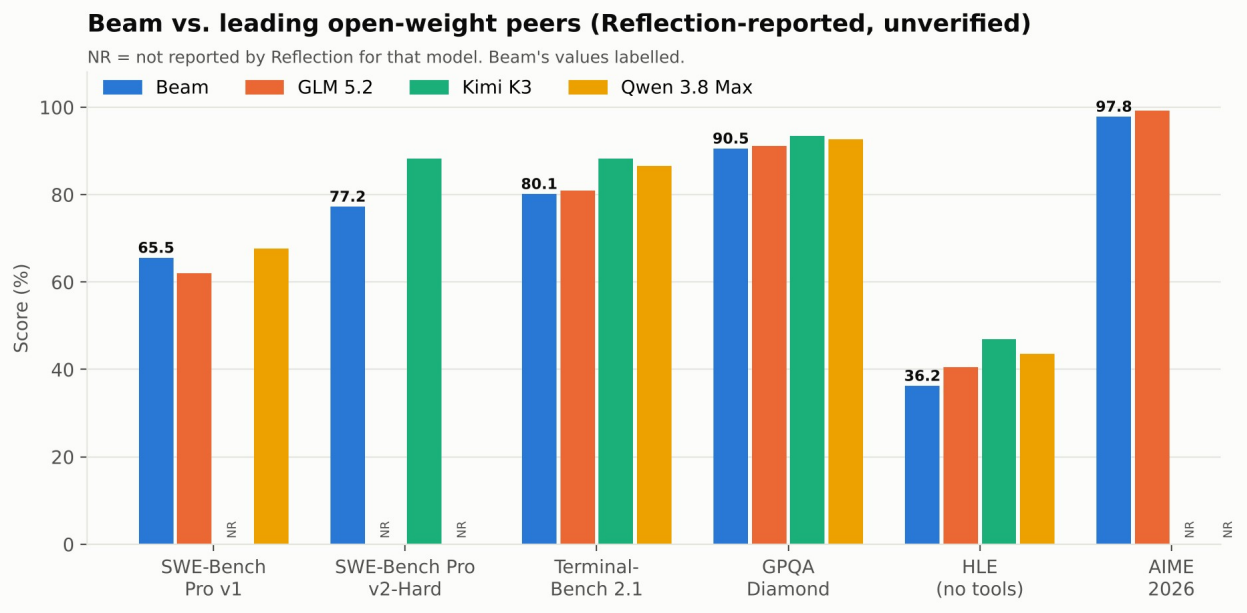


Figure 4 — Reflection's launch table (5 Oct). NR = not reported.

Benchmark	Beam	GLM 5.2	Kimi K3	Qwen 3.8 Max
SWE-Bench Pro v1	65.5 (66.1 card)	62.1	NR	67.7
SWE-Bench Pro v2-Hard	77.2	NR	88.2	NR
SWE-Bench Verified	80.9 (81.6 card)	NR	NR	NR
Terminal-Bench 2.1	80.1	81.0	88.3	86.6
GPQA Diamond	90.5 (90.6 card)	91.2	93.5	92.6

Benchmark	Beam	GLM 5.2	Kimi K3	Qwen 3.8 Max
HLE (no tools)	36.2	40.5	46.9	43.6
AIME 2026	97.8	99.2	NR	NR

8.3 Independent: Artificial Analysis Intelligence Index

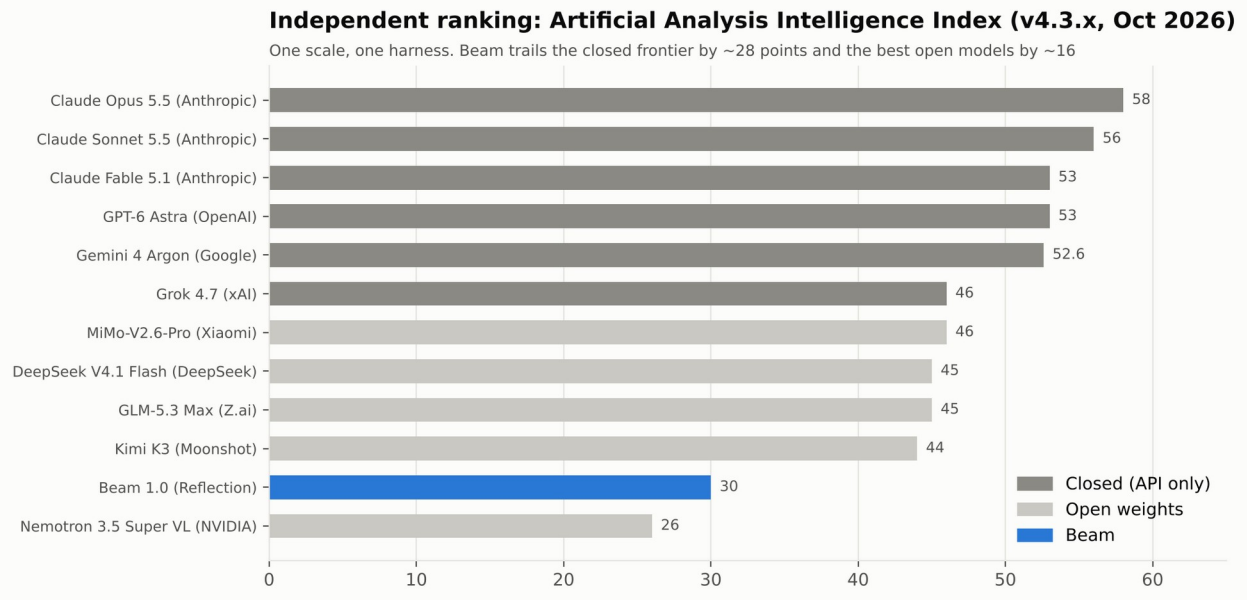


Figure 5 — The only single-scale, single-harness comparison across all vendors.

The independent index tells a different story from the vendor table. Beam (30) ranks 19th of 120 models — above the median for open models of its size and ahead of NVIDIA's Nemotron 3.5 Super VL (26) — but 14–16 points behind the leading Chinese open-weight models and 23–28 points behind the closed frontier. Independent reviewers also note that some Reflection-reported agentic numbers (e.g., AutomationBench) were not reproduced. One clear strength: Beam used ~79M output tokens on the index suite versus a comparison median of ~130M, i.e. it reaches its answers with ~40% fewer tokens.

8.4 Vendor-by-vendor comparison

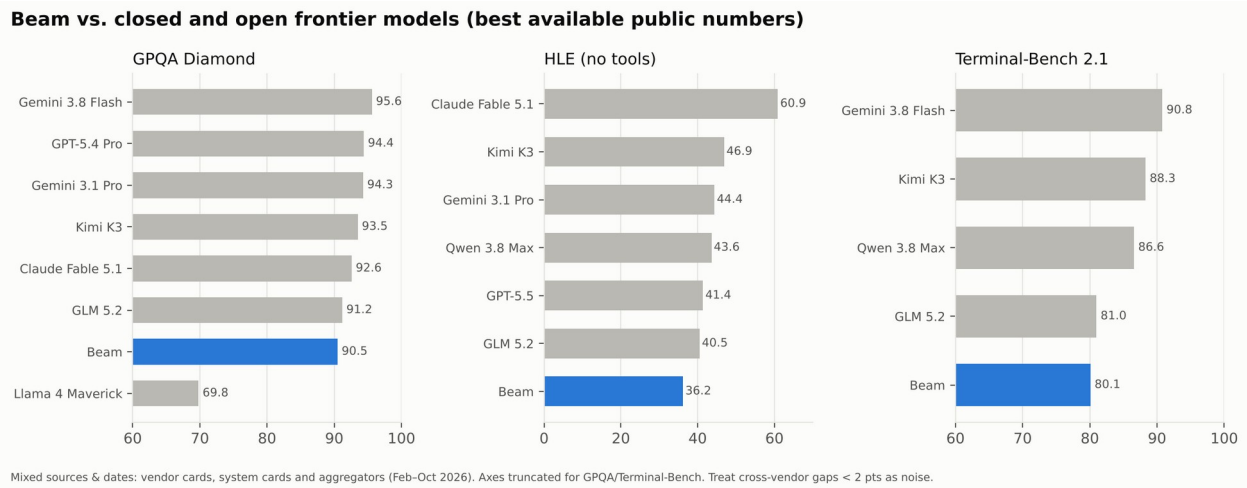


Figure 6 — Best available published numbers on three common benchmarks.

Vendor / model	Weights	AA Index [I]	Headline published results	List price in/out per 1M
Anthropic — Claude Opus 5.5 (Sep 2026)	Closed	58 (Max)	SWE-bench Pro 89.9 [V]; Terminal-Bench 4.0 66.4 [V] (59.6–61.6 [I]); HLE 64.4 / 67.7 with tools [V]	$4 / $20
Anthropic — Claude Sonnet 5.5	Closed	56	Terminal-Bench 4.0 70.6; SWE-bench Pro 81.3 [V]	$2 / $10
Anthropic — Claude Fable 5.1	Closed	≈53	SWE-bench Pro 81.2; HLE 60.9; GPQA 92.6 [V/S]	$10 / $50
OpenAI — GPT-6 Astra	Closed	≈53	Terminal-Bench 4.0 ~58–60 [I/S]; GPQA 96.1 [S]	$10 / $50
OpenAI — GPT-6.1 Sol	Closed	—	Terminal-Bench Science 57.0; OSWorld 2.0 71.4 [S]	$2 / $10
Google — Gemini 4 Argon	Closed	52.6	Roughly level with GPT-6 Astra on the index [I]	—
Google — Gemini 3.8 Flash	Closed	—	GPQA 95.3–95.6; Terminal-Bench 2.1 89.4–90.8; SWE-bench Pro 61.6 [S]	$0.75 / $3.75
xAI — Grok 4.7	Closed	46	Terminal-Bench 4.0 37.6 [V] vs 26 [S, independent]; GPQA (4.6) 94.9 [S]	$2 / $6
Meta — Muse Spark 1.3	Closed	61 on v4.1.1 (different index version)	Terminal-Bench 2.1 88.8 [V]; Meta's self-reports have exceeded verified scores before	—
Meta — Llama 4 Maverick	Open (Llama licence)	—	GPQA 69.8 [V]; Meta's latest open generation	—
Moonshot — Kimi K3	Open	44	Terminal-Bench 2.1 88.3; GPQA 93.5; HLE 46.9 [V via Reflection]	—
Z.ai — GLM-5.3 (Max)	Open	45	GLM-5.2: Terminal-Bench 2.1 81.0; GPQA 91.2 [V]	—
DeepSeek — V4.1 Flash	Open	45	1.6T/49B (V4 Pro) [S]	$0.14 / $0.28 (V4 Flash)
Xiaomi — MiMo-V2.6-Pro	Open	46	Top open model on the index [I]	—
Reflection — Beam 1.0	Open (Apache 2.0)	30	SWE-bench Verified 81.6; Terminal-Bench 2.1 80.1; GPQA 90.6 [V]	Partner-dependent

Notes on specific vendors

- **xAI (Grok):** xAI (now operating as SpaceXAI per some reports) competes primarily on price ($2/$6) and speed; its agentic coding scores lag the leaders, and vendor vs independent Terminal-Bench numbers differ by ~12 points.
- **Meta:** Meta has split its line: the Muse series (Spark, Wonder) is closed and powers Meta AI; Llama 4 remains the latest open generation. A Beam-class open model competes with Llama, not Muse.
- **Groq (the inference company, distinct from xAI's Grok):** does not train frontier models; it hosts open models on LPU hardware and is a potential deployment partner. NVIDIA licensed its technology in Dec 2025 (~US$20B).

8.5 Efficiency: capability per active parameter

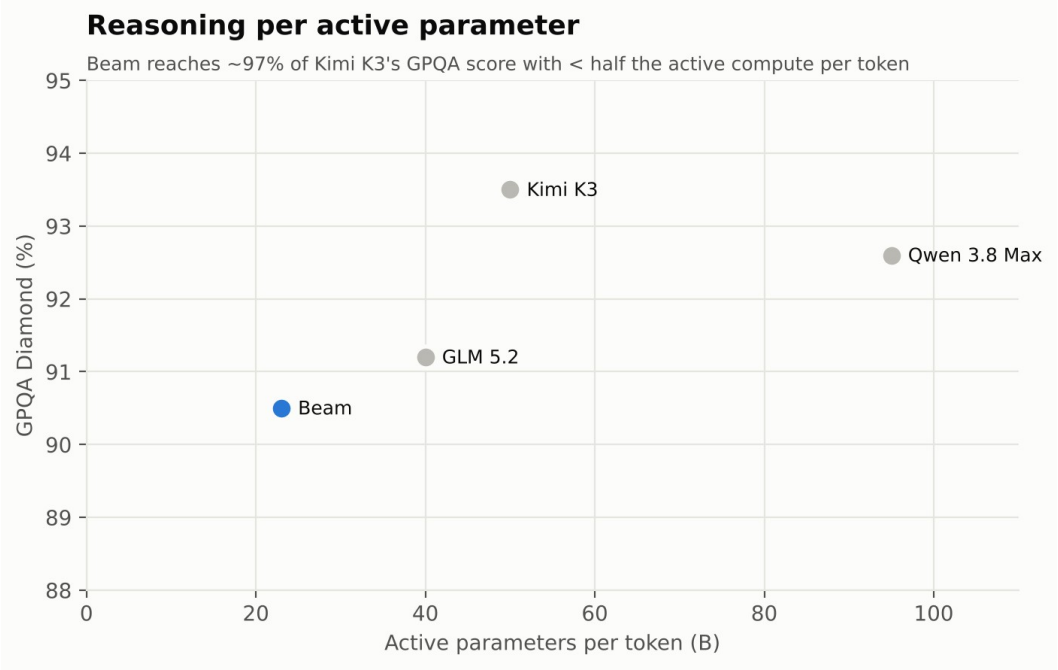


Figure 7 — GPQA Diamond vs. active parameters (vendor-reported).

On vendor numbers Beam reaches ~97% of Kimi K3's GPQA score with under half its active parameters. Independent results narrow the efficiency story to serving economics: Beam's 23.4B active parameters, sparse global attention, NVFP4 checkpoint and drafter make it among the cheapest models of its size to run per token.

9. Market and competitive landscape

9.1 Market size and structure

Indicator	Value	Source / note
Enterprise generative-AI spend	US$37B in 2025, up from US$11.5B (≈3.2×)	[S] Menlo Ventures
Global LLM market (narrow definition)	US$10.6–12.8B in 2026	[S] market-research aggregators; definitions vary
Total AI spending (all categories)	≈ US$2.5T in 2026 (+47% y/y)	[S] Gartner via aggregators
Enterprise API share — Anthropic / OpenAI	40% / 27%	[S] Menlo Ventures, late 2025
U.S. businesses paying (Aug 2026)	Anthropic 43.8%, OpenAI 39.8%	[S] Ramp transaction data
Open-weight share of enterprise workloads	11–13%, down from 19%	[S] Menlo Ventures

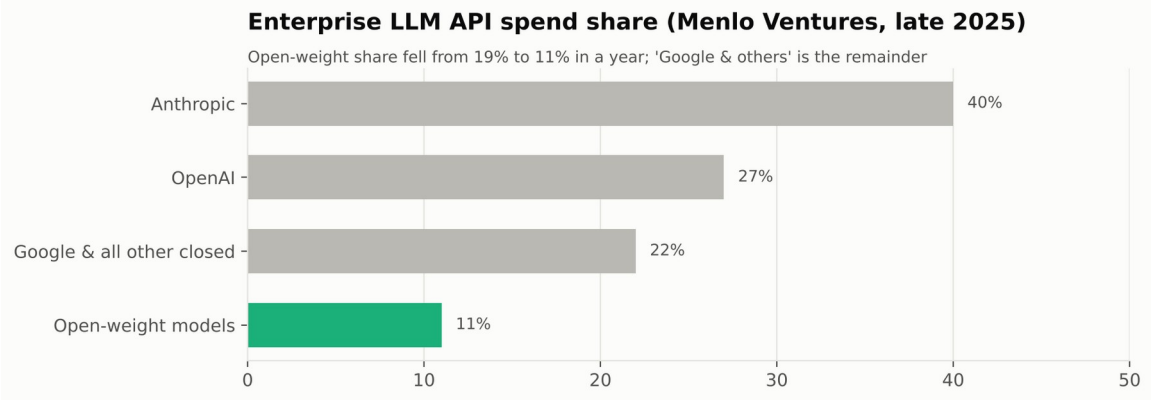


Figure 8 — Enterprise LLM API spend share.

Implication: enterprise demand is concentrated on the most capable closed models, and open-weight share has been falling. An open model must therefore win on dimensions that closed APIs cannot match — control, privacy, residency, customisation and cost — rather than on leaderboard rank alone.

9.2 Price vs. capability

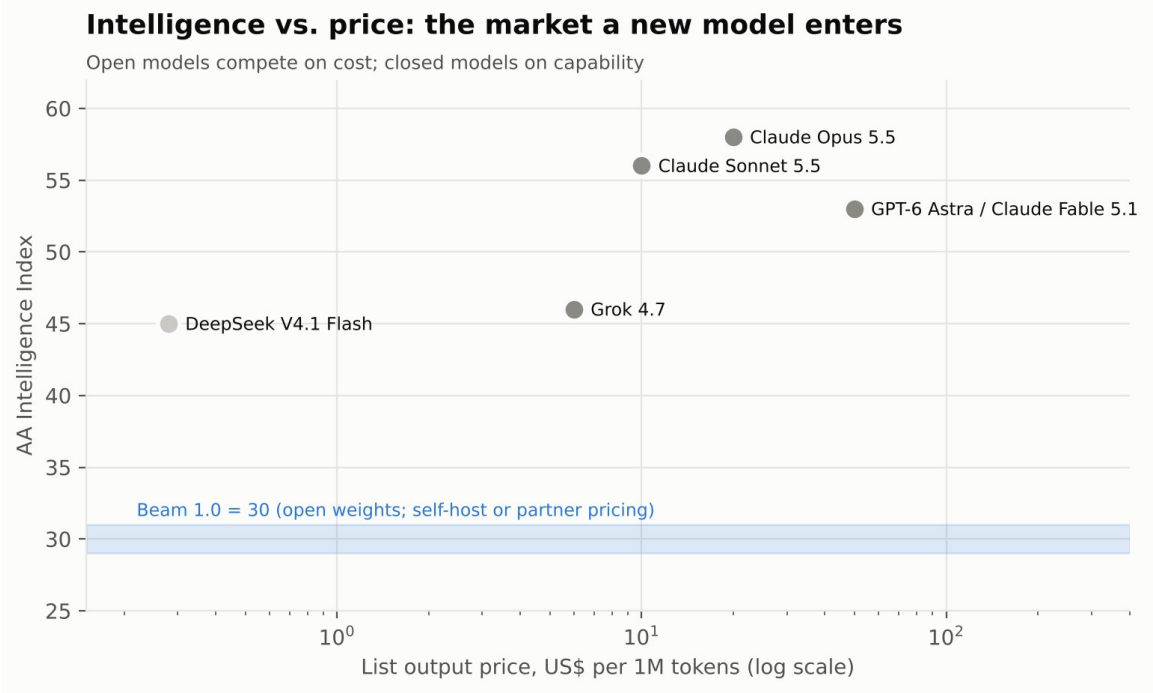


Figure 9 — List output price vs. independent intelligence index.

Model	Input $/1M	Output $/1M	Positioning
Claude Fable 5.1 / GPT-6 Astra	10	50	Premium frontier
Claude Opus 5.5	4	20	Highest independent index
Claude Sonnet 5.5 / GPT-6.1 Sol	2	10	Mid-price workhorse
Grok 4.7	2	6	Price-led challenger
Gemini 3.8 Flash	0.75	3.75	Fast/cheap tier
Claude Haiku 5.5	0.10	0.50	Lowest-cost production tier
DeepSeek V4 Flash	0.14	0.28	Open-weight price floor

Prices from third-party aggregators as of 2–17 Oct 2026; verify against vendor rate cards.

9.3 Competitor profiles

Competitor	Strengths	Weaknesses / opening for a new entrant
Anthropic	Top independent scores; leading enterprise share; strongest agentic coding	Closed; premium pricing; no on-prem weights
OpenAI	Broadest consumer reach; strong reasoning; wide model range	Closed; enterprise share below Anthropic
Google	Multimodal, long context, cheapest fast tier, cloud distribution	Closed weights for Gemini; Gemma open models smaller
xAI (Grok)	Low prices, speed, X distribution	Lags on agentic coding; vendor/independent score gaps
Meta	Massive distribution; Llama brand in open	Frontier Muse models closed; Llama 4 ageing

Competitor	Strengths	Weaknesses / opening for a new entrant
	source	
Chinese open labs (DeepSeek, Moonshot, Z.ai, Alibaba, Xiaomi)	Best open-weight scores; very low prices; rapid release cadence	Geopolitical and procurement restrictions in US/EU government and regulated sectors
Reflection (Beam)	US-built, Apache 2.0, efficient, strong ecosystem launch	Independent score well below Chinese open leaders; full report pending
Mistral, NVIDIA (Nemotron)	European / hardware-vendor open models	Lower independent scores; narrower focus

9.4 Where a new Beam-class model can win

1. **Sovereign and government AI:** US/EU/allied governments that cannot procure Chinese open weights and need on-prem deployment.
2. **Regulated industries:** finance, healthcare, defence and legal, where data residency and auditability outweigh a few index points.
3. **Vertical specialists:** a model post-trained on proprietary domain environments (e.g., chip design, cybersecurity, scientific code) can beat general frontier models on narrow tasks.
4. **Cost-sensitive agentic workloads at scale:** high-volume coding agents and background automation where per-token cost dominates.
5. **Language and regional markets:** under-served languages (e.g., Hebrew/Arabic, regional European languages) where targeted data yields outsized gains.

10. Gap analysis: what it takes to compete

Tier	Representative models	AA Index	Typical scale	Gap from Beam
Closed frontier	Claude Opus/Sonnet 5.5, GPT-6 Astra, Gemini 4 Argon	53–58	Undisclosed; likely multi-trillion parameters and 10^25–10^26 FLOPs	+23 to +28
Price-led closed	Grok 4.7	46	Undisclosed	+16
Open frontier	MiMo-V2.6-Pro, GLM-5.3, DeepSeek V4.1, Kimi K3	44–46	0.75–2.8T total; 40–95B active	+14 to +16
Beam class	Beam 1.0	30	501B / 23B	—

10.1 Levers that close the gap (in order of expected return)

1. **Post-training scale and environment diversity:** more RL compute, more verifiable environments, better reward design.
2. **Active-parameter budget:** the open leaders use 40–95B active vs Beam's 23B; doubling active parameters is the most direct capability lever, at proportionally higher serving cost.
3. **Data quality and quantity:** 30T+ curated tokens with larger synthetic reasoning/code shares.
4. **Total capacity:** 1–2T total parameters with the same active budget raise knowledge capacity without raising per-token compute.
5. **Evaluation-driven iteration:** independent-harness evaluation each checkpoint to avoid optimising vendor-specific harnesses.

11. KPI framework

“PKI” in the original brief is interpreted as KPI — key performance indicators. Three KPI levels are defined. All must be measured on a fixed, independent harness, and Beam itself (now downloadable) must be re-run on the same harness as the baseline.

KPI	Level 1: Beam parity	Level 2: Open-frontier	Level 3: Frontier challenger
AA Intelligence Index (or equivalent composite)	≥ 30	≥ 45	≥ 53
SWE-bench Verified	≥ 81	≥ 85	≥ 90
SWE-bench Pro (v1 / public)	≥ 66	≥ 72	≥ 85
Terminal-Bench 2.1	≥ 80	≥ 87	≥ 90
GPQA Diamond	≥ 90	≥ 93	≥ 95
HLE (no tools)	≥ 36	≥ 45	≥ 60
AIME (current year)	≥ 97	≥ 99	≥ 99
Active / total parameters	≤ 25B / ≤ 550B	≤ 50B / ≤ 1.5T	Open
Effective context (RULER ≥ 85%)	≥ 256K (1M stretch)	≥ 1M	≥ 1M
Output tokens per task vs. median	≤ 0.65×	≤ 0.65×	≤ 0.75×
Quality loss at NVFP4	≤ 2 pts	≤ 1.5 pts	≤ 1 pt
Expert load imbalance (max/mean)	≤ 1.10×	≤ 1.05×	≤ 1.05×
Serving cost per 1M output tokens (self-hosted) [E]	≤ US$0.60	≤ US$1.00	≤ US$3.00
Safety	Pass published red-team + bio/cyber uplift evals	Same + third-party audit	Frontier safety framework

12. Strategic options and cost model

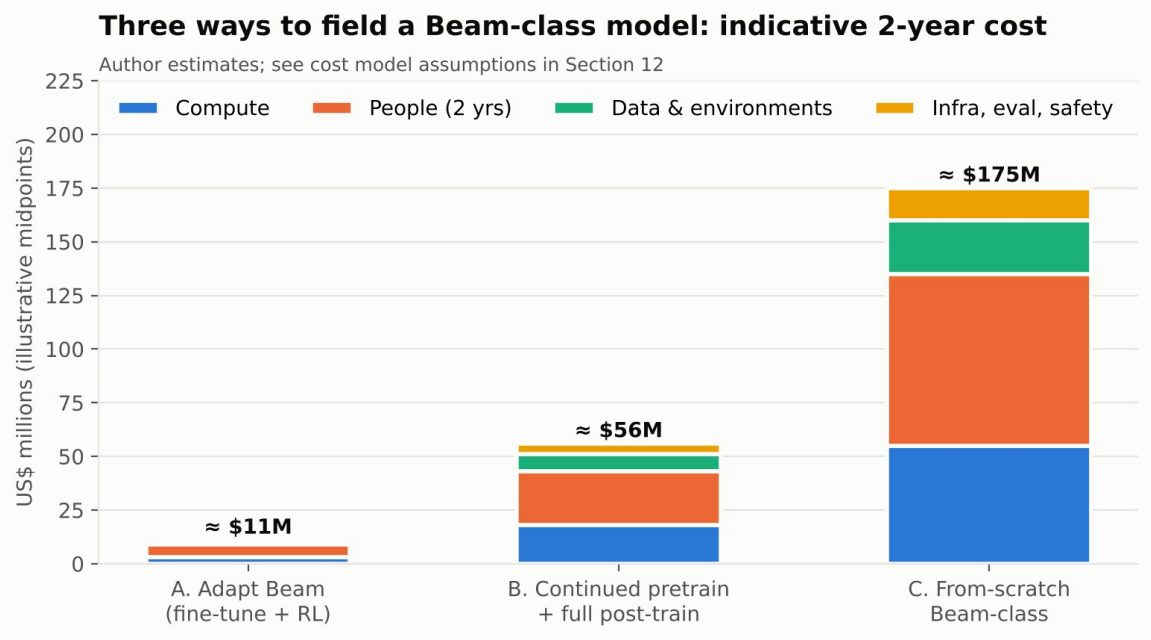


Figure 10 — Indicative two-year cost of three routes to a Beam-class model.

	A. Adapt Beam	B. Continued pre-train + full post-train	C. From-scratch Beam-class
What	SFT + RL on Beam weights for a target domain/language; own evaluation & serving	Add 3–6T domain tokens to Beam, extend context, run Beam-scale RL	Own architecture, tokenizer, 24T+ token pre-train, full RL
Time to first model	3–6 months	9–12 months	15–18 months
Compute [E]	≈ 0.5–1M GPU-hrs (≈ US$3M)	≈ 4–6M GPU-hrs (≈ US$18M)	≈ 10–14M GPU-hrs (≈ US$55M)
Team	15–25	40–60	80–120
Indicative 2-yr total	≈ US$11M	≈ US$56M	≈ US$175M
IP ownership	Derivative of Apache 2.0 weights (permissive)	Derivative, substantially modified	Fully owned
Ceiling	Beam + domain gains	Beam + 5–10 index points in domain	Can scale to Level 2 with more capital
Main risk	Limited differentiation	Catastrophic forgetting; data cost	Execution risk; capital

12.1 Compute model for option C [E]

Phase	Estimate	Assumptions
Pre-training	≈ 3.3 × 10^24 FLOPs → ≈ 2–3M GB300-hours	6 × 23.4B × 23.8T; 20–30% MFU on ~2.5 PFLOP/s BF16-dense (higher with FP8)
Mid-training	≈ 0.2–0.4M GPU-hours	~1T tokens at increasing sequence length
RL	≈ 7M GPU-hours	10,500 GPUs × 28 days (Beam's disclosed scale)

Phase	Estimate	Assumptions
Ablations & pilots	≈ 1–2M GPU-hours	1–7B dense/MoE and ~60B-A3B pilots
Inference for evals & synthetic data	≈ 0.5–1M GPU-hours	Data generation, LLM-judge, eval sweeps
Total	≈ 11–14M GPU-hours	≈ US$40–70M at US$3–5 per GPU-hour

12.2 Team composition for option C

Function	Headcount	Key roles
Pre-training & architecture	15–20	Scaling-law research, MoE kernels, distributed training
Data	15–25	Crawling, filtering classifiers, synthetic data, licensing
Post-training & RL	15–25	RL algorithms, reward modelling, environment builders
Infrastructure	15–20	Cluster ops, storage, networking, sandbox fleet, inference
Evaluation & safety	8–12	Benchmarks, red-teaming, policy
Product, GTM, partnerships	10–15	Developer relations, enterprise sales, cloud partners
Total	≈ 80–120	Fully loaded cost ≈ US$300–450K per head-year in the US [E]

13. Reference architecture for the new model

The proposed design adopts Beam's confirmed choices as defaults and tests a small number of improvements suggested by 2026 research. All non-default choices must win in ablations at 1–7B scale and a ~60B-A3B pilot before full training.

Component	Default (Beam-matched)	Candidate improvement to ablate	Rationale
Size	≈ 500B total / ≈ 23B active, 52 layers	≈ 600B / 30B active	Most direct capability lever; still fits 8-GPU node in NVFP4
Dense layers	First 3 layers dense	1–4 dense	Early-layer stability
Experts	256 routed, top-8	+1 shared expert; 384 routed	DeepSeek-V3/V4 practice
Routing	Aux-loss-free bias balancing, cosine-decayed updates	Sequence-level balance regulariser	Reproduce 1.04× load
Attention	Gated GQA; SWA-512 on 43 layers; 9 NoPE global layers	Compressed/sparse global attention (DeepSeek-V4 style); MLA on global layers	Further KV reduction at 1M context
Norm / residual	SandwichNorm; depth-scaled residual; FP32 residual accumulation	+ QK-norm	Low-precision stability
Tokenizer	≈ 152K BPE	≈ 200K with target-language coverage	Compression on code and target languages
Prediction head	Next-token	Multi-token prediction (1–2 heads)	Denser signal; built-in speculative decoding
Optimiser	AdamW (assumed)	Muon for matrix params (per-expert), AdamW for others	~2× token efficiency reported
Precision	BF16/FP8 training	NVFP4 training on GB300	≈ 2× matmul throughput
Drafter	DFlash2-style ~2B drafter	MTP-head self-drafting	Latency for agent loops

14. Data strategy

14.1 Pre-training corpus target (≈ 24–30T tokens) [E]

Source	Share	Notes
Filtered web text	40–50%	Model-based quality classifiers; keep top ~5–10% (Beam discarded ~95%)
Code (permissive licences)	20–25%	Repository-level packing; dedup against benchmarks
Mathematics & science	8–12%	Papers, textbooks, formal proofs
Synthetic reasoning / code	10–15%	Execution-verified solutions, rephrased high-quality text
Multilingual & target languages	5–10%	Up-weight target markets (e.g., Hebrew, Arabic, EU languages)
Books, reference, licensed data	3–5%	Licensed where required

14.2 Data engineering requirements

- Exact and fuzzy deduplication (MinHash/LSH) at document and repository level.
- Benchmark decontamination against every KPI benchmark, with n-gram and embedding checks.
- PII removal, toxicity filtering, licence tracking and opt-out handling (copyright and EU AI Act transparency).
- Data-mixture ablations on 1–7B proxies; annealing-phase mixture with the highest-quality data.
- Long-context corpus: repositories, books and multi-document sets for mid-training to 1M tokens.

15. Post-training and agentic RL

1. **SFT:** high-quality instruction, tool-use and agent-trace data in the target harness format (Mirror CLI-compatible).
2. **Environment fleet:** target 10^5–10^6 verifiable environments — unit-test-graded repositories, terminal tasks, maths/science checkers, web and tool tasks — with sandbox orchestration able to run billions of executions (Beam used ~1.3B sandboxes).
3. **RL algorithm:** GRPO-family policy optimisation with asynchronous rollout generation, importance-ratio clipping for off-policy correction, and 128K–256K rollout contexts.
4. **Reward design:** prefer verifiable rewards; use reward models only where verification is impossible; guard against reward hacking with held-out graders.
5. **Token efficiency:** add length-aware rewards to keep output tokens per task ≤ 0.65× the market median (Beam's distinctive strength).
6. **Safety training:** refusal calibration, cyber/bio uplift mitigations and system-prompt robustness before release.

16. Infrastructure plan

Component	Requirement (option C) [E]	Notes
Training cluster	≈ 6,000 GB300-class GPUs for ~1 month (pre-train); ≈ 10,000 for ~1 month (RL)	Rent from neoclouds/hyperscalers; reserve 9–12 months ahead
Interconnect	NVLink 72-GPU domains; 800G InfiniBand/Ethernet scale-out	Expert parallelism is all-to-all heavy
Storage	≈ 10–20 PB for raw and processed data; fast checkpoint store	Checkpoints ≈ 6 TB each with optimiser state
Sandbox fleet	CPU clusters / containers for RL environments (millions of concurrent sandboxes at peak)	Often the hidden bottleneck
Inference for RL	Dedicated vLLM/SGLang rollout servers with weight sync	Asynchronous actor–learner design
Serving	8-GPU nodes in NVFP4 with drafter; autoscaling	Partner with clouds / Groq-class inference providers

17. Evaluation plan

- Fixed internal harness covering all Section 11 KPIs, run on every checkpoint with confidence intervals.
- Re-run Beam 1.0, GLM-5.x, Kimi K3 and DeepSeek V4.x on the same harness; never compare against vendor tables alone.
- Prefer unsaturated and contamination-resistant suites: SWE-bench Pro, Terminal-Bench 2.x/4.0, HLE, current-year AIME, RULER, τ-bench/τ^3-bench, OSWorld.
- Submit to Artificial Analysis, Vals AI and LMArena before launch for independent numbers.
- Track token efficiency and cost per solved task alongside accuracy.
- Domain-specific private evals for the chosen vertical; these define commercial success more than public indices.

18. Safety, legal and governance

Topic	Consideration
Licence of base weights	Apache 2.0 permits commercial use, modification and redistribution; preserve NOTICE and attribution; includes a patent grant
EU AI Act	General-purpose AI model obligations (technical documentation, copyright policy, training-data summary). Models above 10^25 training FLOPs are presumed to carry systemic risk — a Beam-class model (≈3–4 × 10^24 [E]) sits below this threshold; a Level 2/3 model likely exceeds it
Copyright and data	Licence tracking, opt-out compliance, provenance records; litigation risk is concentrated in training-data choices
Export controls	Track US rules on model weights and advanced compute; open-weight release decisions need legal review
Safety evaluation	Pre-release red-teaming, cyber and bio uplift evaluation, and a published safety report (Reflection published a Beam Safety Technical Report)
Responsible release	Staged release (API → weights), model card, acceptable-use policy

19. Go-to-market and business model

Model	Description	Revenue logic
Open weights + managed API	Release weights; sell a hosted API and enterprise SLA	Usage-based; low price per token
Sovereign / on-prem licensing	Packaged deployment for governments and regulated enterprises	Annual licence + support; high contract value
Vertical model	Domain-specialised model and agent product	Subscription per seat or per task
Cloud marketplace	List on AWS, Azure, Google Cloud, NVIDIA NIM	Revenue share; distribution leverage
Custom post-training services	Fine-tune/RL on customer environments	Project fees + recurring hosting

Beam's launch is the template: an open-weights release integrated with 40+ partners on day 0, plus an open-source harness that makes developers productive immediately. A new entrant should secure cloud and inference partners before release and ship serving recipes for vLLM and SGLang on day 0.

20. Risks and mitigations

Risk	Likelihood	Impact	Mitigation
Capability gap to frontier widens faster than the plan can close it	High	High	Do not compete head-on; target niches where open weights win
Chinese open models keep a 15-point lead at near-zero price	High	Medium	Differentiate on provenance, sovereignty, support and domain quality
Vendor benchmark claims do not replicate	Medium	High	Independent harness from day one; publish third-party results
GPU availability / price shocks	Medium	High	Multi-provider reservations; NVFP4 to cut compute needs
RL infrastructure underestimated	High	High	Start environment fleet and sandbox orchestration in month 1
Key-person dependence / hiring	Medium	High	Competitive equity; partner with universities
Legal (copyright, export control, AI Act)	Medium	Medium–High	Provenance tracking; legal review gates
Consolidation (e.g., Reflection acquired; Beam licence terms change for future versions)	Medium	Medium	Mirror Apache 2.0 Beam 1.0 weights now; the grant is irrevocable for released versions
Commoditisation of open models	High	Medium	Monetise services, deployment and vertical products rather than raw weights

21. Roadmap and decision gates

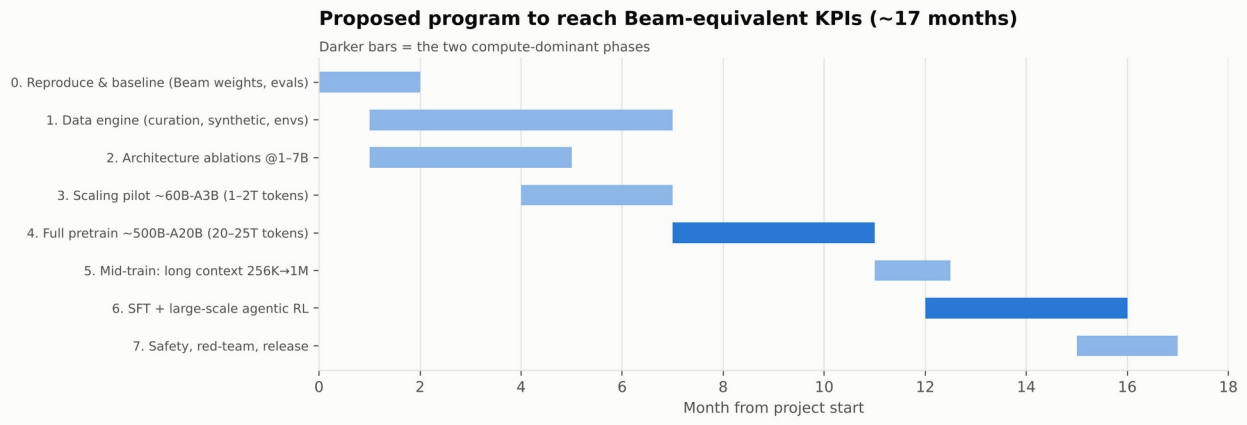


Figure 11 — Programme roadmap for option C; options A and B use phases 0–2 and 5–7.

Gate	Timing	Go criteria
G0 — Baseline	Month 1	Beam 1.0 and peers reproduced on internal harness within ±2 pts of published values
G1 — Adapt	Month 6	Adapted Beam beats base Beam by ≥ 5 pts on domain evals; first paying pilot customer
G2 — Scale decision	Month 9	Pilot (60B-A3B) matches scaling-law predictions; data engine ≥ 20T curated tokens; funding ≥ US$150M secured
G3 — Full pre-train	Month 11	Loss and ablation targets met; no instability in first 10% of tokens
G4 — Release	Month 17	Level 1 KPIs met independently; safety report complete; ≥ 10 launch partners

22. Recommendations from current research

22.1 Architecture

1. **Fine-grained MoE with aux-loss-free balancing** (256 experts, top-8) — confirmed in Beam and DeepSeek-V3/V4.
2. **Aggressively sparse global attention:** Beam shows 9/52 global layers with SWA-512 elsewhere is viable at 1M context; evaluate DeepSeek-V4-style compressed sparse attention for the global layers.
3. **NoPE on global layers** for length generalisation; RoPE (or none) on local layers.
4. **Gated attention, SandwichNorm, QK-norm and FP32 residuals** for low-precision stability.
5. **Multi-token prediction** for sample efficiency and self-speculative decoding (DeepSeek-V3/V4, MiMo-V2.6).

22.2 Optimisation and precision

1. **Muon optimiser** for matrix parameters (Moonlight: ~2× compute efficiency vs AdamW; adopted by DeepSeek-V4 with per-expert orthogonalisation).
2. **FP8 → NVFP4 training and serving:** Beam's NVFP4 checkpoint loses ≤ 1.7 points; plan NVFP4 serving from day one.
3. **WSD learning-rate schedules** to branch mid-training and annealing runs from a shared trunk.

22.3 Data and post-training

1. **Data engine before model:** aggressive quality filtering and execution-verified synthetic data.
2. **RL compute ≥ pre-training compute,** with a million-scale environment fleet and an open harness (Mirror CLI) shared between training and deployment.
3. **Reward token efficiency** explicitly to reproduce Beam's ~40% token saving.

22.4 Strategy

1. **Start from Beam 1.0 weights.** Apache 2.0 makes it the cheapest credible starting point for a US/EU-provenance model.
2. **Measure independently, always.** Beam's gap between vendor and independent numbers is the central lesson of its launch.
3. **Pick one axis to lead on** — sovereignty, a vertical, a language, or cost — rather than the general leaderboard.

23. Download links and access

Beam 1.0 is now downloadable (released 16 October 2026, Apache 2.0). Verify the publisher is Reflection's official Hugging Face organisation ("reflection").

Resource	Location	Notes
Beam 1.0 (BF16)	huggingface.co/reflection/Beam-1.0-501B-A23B	≈ 1 TB
Beam 1.0 (FP8)	huggingface.co/reflection/Beam-1.0-501B-A23B-FP8	≈ 0.5 TB
Beam 1.0 (NVFP4)	huggingface.co/reflection/Beam-1.0-501B-A23B-NVFP4	≈ 0.28 TB; Blackwell
Community GGUF	huggingface.co/unsloth/Beam-1.0-501B-A23B-GGUF	Unofficial quantisations
Hugging Face launch blog	huggingface.co/blog/reflection-beam	Usage guide
vLLM serving recipe	recipes.vllm.ai/reflection/Beam-1.0-501B-A23B	B300 / H200 commands
NVIDIA Dynamo recipe	docs.nvidia.com/dynamo/dev/recipes/beam-1-0-501b	Disaggregated serving
AWS guide	aws.amazon.com/blogs/machine-learning/…reflection-beam…	Agentic coding on AWS
Google Cloud guide	cloud.google.com/blog/…reflection-beam…	Day-0, 5× performance claim
Mirror CLI, DFlash2 drafter, Safety Technical Report	Linked from Reflection's "Beam is now available" post	Open-source harness and drafter
Open-weight peers	huggingface.co/deepseek-ai · /moonshotai · /zai-org · /Qwen · /mistralai · /meta-llama · /openai	Baselines for comparison
Closed APIs	console.anthropic.com · platform.openai.com · aistudio.google.com · console.x.ai	API only
Inference providers	console.groq.com and others	Hosted open models

Clickable links:

- Beam 1.0 BF16 on Hugging Face
- Beam 1.0 FP8 on Hugging Face
- Beam 1.0 NVFP4 on Hugging Face
- Unsloth GGUF quantisations
- Hugging Face blog: Welcome Beam
- Reflection: Beam is now available
- vLLM recipe
- NVIDIA Dynamo recipe
- AWS: Reflection Beam for agentic coding
- Google Cloud: day-0 support for Beam
- Artificial Analysis: Beam article
- DeepSeek on Hugging Face
- Moonshot AI (Kimi) on Hugging Face
- Z.ai (GLM) on Hugging Face
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- Meta Llama on Hugging Face

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Appendix A. Formulas and worked estimates

Quantity	Formula	Beam example [E]
Training FLOPs	≈ 6 × N_active × D_tokens	6 × 23.4e9 × 23.8e12 ≈ 3.3 × 10^24
GPU-hours	FLOPs ÷ (peak FLOP/s × MFU × 3,600)	3.3e24 ÷ (2.5e15 × 0.25 × 3,600) ≈ 1.5M (BF16); range 1.5–3M
Decode FLOPs per token	≈ 2 × N_active	≈ 47 GFLOPs per output token
Weight memory	N_total × bytes per parameter	501B × 2 B ≈ 1.0 TB (BF16); × 1 B ≈ 0.5 TB (FP8); × ~0.56 B ≈ 0.28 TB (NVFP4)
KV cache share at long context	global layers ÷ total layers	9 ÷ 52 ≈ 17% of an all-global design
Active ratio	N_active ÷ N_total	23.4 ÷ 501.3 ≈ 4.7%

Appendix B. Glossary

Term	Meaning
MoE	Mixture-of-Experts: many expert sub-networks per layer, few active per token
501B-A23B	501B total parameters, ~23B active per token
Top-8 of 256	Router selects 8 of 256 experts for each token
SWA	Sliding-window attention: attends only to the last N tokens (512 in Beam)
NoPE	No positional encoding on a layer
GQA / MLA	Grouped-query / multi-head latent attention: KV-cache reduction techniques
SandwichNorm	Normalisation before and after each sub-layer
NVFP4	NVIDIA 4-bit floating-point format on Blackwell GPUs
Speculative decoding / drafter	A small model proposes tokens that the large model verifies in parallel
MTP	Multi-token prediction heads
Muon	Optimiser that orthogonalises momentum updates for matrix parameters
MFU	Model FLOPs utilisation
AA Intelligence Index	Artificial Analysis composite of reasoning, knowledge, coding and agentic evaluations
GPQA / HLE / AIME	Graduate science QA / Humanity's Last Exam / maths olympiad qualifier
SWE-bench / Terminal-Bench / τ-bench	Software-engineering, terminal-agent and tool-use agent benchmarks
KPI	Key performance indicator